

		Steel Wire Incoming Inspection - Registration					Date: 5/17/25
Characteristic Product/Process	Tolerance of specification/process	Size (pcs)	Frequency	Measurement technique	Control method	Initial/record for Verification of findings	
Correct steel type, coils quantity and supplier tag presence	Steel type and q.ty according delivery doc. and certificate	100%	Each arrival	Visual	Check, stamp and sign the delivery certificate.	Ok <i>OK</i> Nok	
Chemical, metallurgical and mechanical characteristics	ST 070403	100%	Each arrival	Visual	Check supplier test report. Stamp and sign	Ok <i>OK</i> Nok	
Chemical characteristic	ST 070403	1	Every new heat	Quantometer	Variables control. Report saved on PC	Ok Nok	
Surface quality (scratches, pits, slivers, laps, coil strapping, etc.)	ST 070401 ST 070403	100%	Each arrival	Visual	Check the appearance of the surface of the wire.	Ok <i>OK</i> Nok	
Wire diameter	ST 070401	100%	Each coil	Vernier caliper	Check the diameter on multiple position (10) on every single coil.	Ok <i>OK</i> Nok	
Decarburization	ST 070403	1	Every new heat	Metallographic microscope	Quarta 3 - Preload Raw material	Ok <i>OK</i> Nok	
Presence of phosphating	QCWI 021	1	Each Coil	Visual	Check the appearance of the surface of the wire, compare to pictures	Ok <i>OK</i> Nok	

Name	LS	OEB Internal Code #	021296
Date	5/17/25	BOL #	11814707
Heat #	10912380	Charter Steel Order #	10392533
Finish Size	10.40mm 10B30	Number of Coils	5
		Sign <i>[Signature]</i>	

STRAIGHT BILL OF LADING-SHORT FORM-THE SHIPPING ORDER

RECEIVE, subject to the classification and tariffs in effect on the date of issue of this shipping order.

The property described below, in apparent good order, except as noted (contents and conditions of contents of packages unknown), marked, consigned, and destined as indicated below, which said carrier (the word carrier begin understood throughout this contract as meaning any person or corporation in possession of the property under the contract) agrees to carry to its usual place of delivery at said destination, if on its route, otherwise to deliver to another carrier on the route to said destination. It is mutually agreed, as to each carrier of all or any of said property over all or any portion of said route to destination, as to each party at any interested in all or any of said property, that every service to be performed hereunder shall be subject to all the terms and conditions of the Uniform Domestic Straight Bill of Lading set forth (1) in Uniform Freight Classification in effect on the date hereof, if this is a rail or rail-water shipment, or (2) in the applicable motor carrier classification or tariff if this is a motor carrier shipment. Shipper hereby certifies that he is familiar with all the terms and conditions of the said bill of lading, including those on the back thereof, set forth in the classification or tariff which governs the transportation of this shipment, and the said terms and conditions are hereby agreed to by the shipper and accepted for himself and his assigns.

NO GOODS WILL BE ACCEPTED
FOR RETURN WITHOUT OUR
WRITTEN CONSENT. CLAIMS FOR
LOST OR DAMAGED GOODS
SHOULD BE MADE WITHIN 5 DAYS
OF RECEIPT OF GOODS.

FROM: CHARTER STEEL**AT: Saukville, WI 53080****Shipper's No.** _____**Agent's No.** _____

Or rail water shipment, or (2) in the applicable motor carrier shipment.

CHARTER STEEL ORDER #: 10392533 1.1**B
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O**

Brugola OEB Industriale USA
45555 Port St
Plymouth Township , MI-
48170,US

**BOL****S
H
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P
T
O**

Brugola OEB Industriale USA
45555 Port St.
Plymouth Twp , MI-48170,US

MADE IN USA

TRIP No.	DELIVERY No.	BILL OF LADING No.	VERSION No.	CUST No.	SHIPPED VIA		CLASS
R6630896	2600262	11814707		8343	JD LOGISTICS TRUCKING		50
SHIP DATE (CST) Central time	CAR TRAILER ID	TRAILER TYPE	INITIALS	FREIGHT TERMS	No. OF CARRIERS	SIZE OF CARRIERS	CHARTER PART
05-MAY-2025 18:00	5490	Van		Due	5	YELLOW 35" X 58" X 58"	372704

CUST PO: 4515083070

CUST PART	DESCRIPTION	NON CHARTER HEAT	SHIP LOT	CHARTER HEAT	LOT No.	WT/LBS
9940071040	PK*10B30*R*SK*FG*RHQ*2 9/64*RND*COIL*DFSAR*MPH OSPOLY*0.409*4400		4886492 4892139 4884598 4886492 4892139	10912380 10912380 10912380 10912380 10912380	107-596582-00 107-596578-00 107-596577-00 107-596581-00 107-596580-00	4330 ✓ 4329 ✓ 4373 ✓ 4381 ✓ 4361 ✓

Subtotal 21774**Carrier Weight:****390****Dispatch Quantity:****21774****Total Dispatch Quantity:****22164****Order Notes :****Shipping Instructions:**

All carriers are property of Charter Steel. For carrier returns: 800-789-7155 or carriers@chartersteel.com
Load Configuration: Eye to Rear
Green Trailer Type: Van
Process Trailer Type: Van
Tarping Requirement: Not Required
ALL COILS SHOULD BE ON CARRIERS

Total Number of 5**INSPECTED****BY** _____ **DATE** _____**Truck Driver's Signature**

Fumes or particulars released from metal working activities like welding, grinding, heating or similar activities can be dangerous to your health. Provide ventilation. If breathing is difficult or stops, administer artificial respiration or oxygen as required. Seek medical aid. Refer to MSDS for details.

Notice: Freight moving under this bill of lading is subject to the classifications and lawfully filed tariffs in effect on the date of this bill of lading. This notice supersedes and negates any claimed alleged or asserted oral written contract, promise, representation or understanding between the parties with respect to the freight, except to the extent of any written contract which establishes lawful contract carriage and is signed by authorized representatives of both parties to the contract. All carriers are property of Charter Steel. For carrier returns: 800-789-7155

Picked up by: _____**Electronic Copy****Received By:** _____



**CHARTER
STEEL**

A Division of
Charter Manufacturing Company, Inc.

FILE

1658 Cold Springs Road
Saukville, Wisconsin 53080
(262) 268-2400
1-800-437-8789
Fax (262) 268-2570

CHARTER STEEL TEST REPORT

Melted in USA Manufactured in USA

Brugola OEB Industriale USA
45555 Port Street
Plymouth Twp, MI-48170

Cust P.O.	4515083070
Customer Part #	9940071040
Charter Sales Order	10392533
Heat #	10912380
Ship Lot #	4892139
Grade	10B30 R SK FG RHQ 29/64 RNDCOIL
Process	DFSAR
Finish Size	0.409
Ship date	05-MAY-25

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed below and that it satisfies these requirements. The recording of false, fictitious and fraudulent statements or entries on this document may be punishable as a felony under federal statute.

Test results of Heat Lot # 10912380

Lab Code: 7388

CHEM	C	MN	P	S	SI	NI	CR	MO	CU	SN	V
%Wt	.29	.84	.007	.009	.210	.04	.08	.02	.08	.006	.003
	AL	N	B	TI	NB						
	.025	.0070	.0027	.020	.001						

JOMINY(HRC)

J1	J2	J3	J4	J5	J6	J7	J8	J9
51	50	49	46	34	24	21	20	20
J01.5MM	J03MM	J05MM	J07MM	J09MM	J11MM	J13MM	J15MM	
50	49	47	42	33	27	24	21	

JOMINY LAB=0358-01

JOMINY SAMPLE TYPE METRIC=C

JOMINY SAMPLE TYPE ENGLISH=R

E45 INCLUSION LAB=0358-02

	A	B	C	D							
Thin	1.0	1.4	.0	.6							
Heavy	.2	.2	.0	.5							
LOC1=1-6	AT=1.0	AH=.0	BT=1.5	BH=.5	CT=.0	CH=.0	DT=.5	DH=.5			
LOC2=1-11	AT=1.0	AH=.0	BT=1.0	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			
LOC3=1-12	AT=1.0	AH=.5	BT=2.0	BH=.0	CT=.0	CH=.0	DT=1.0	DH=.5			
LOC4=2-11	AT=1.0	AH=.5	BT=1.5	BH=.5	CT=.0	CH=.0	DT=.5	DH=.5			
LOC5=3-11	AT=1.0	AH=.0	BT=2.0	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			
LOC6=4-11	AT=1.0	AH=.0	BT=.5	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			

Test results of Rolling Lot # 1409828

REDUCTION RATIO=188:1

		Test results of Processing Lot #4882282, 4892139			TENSILE LAB - Metric = 0358-07 RA LAB - Metric = 0358-07
TENSILE (MPa)	# of Tests	Min Value	Max Value	Mean Value	
	1.0	580	580	580	
REDUCT. AREA - Metric (%)	1	62	62	62	

NUM DECARB=2

FREE FERRITE DECARB (Inch)=.000

FREE FERR & PARTIAL DECARB (Inch)=.004

Melt Source:

This MTR supersedes all previously dated MTRs for this order

Charter Steel

Saukville, WI, USA

Douglas Jones Division Mgr. of Quality Assurance

jonesdo@chartersteel.com

Trip: R6630896

Printed Date : 05/05/2025



**CHARTER
STEEL**

A Division of
Charter Manufacturing Company, Inc.

FILE

1658 Cold Springs Road
Saukville, Wisconsin 53080
(262) 268-2400
1-800-437-8789
Fax (262) 268-2570

CHARTER STEEL TEST REPORT

Melted in USA Manufactured in USA

Brugola OEB Industriale USA
45555 Port Street
Plymouth Twp, MI-48170

Cust P.O.	4515083070
Customer Part #	9940071040
Charter Sales Order	10392533
Heat #	10912380
Ship Lot #	4892139
Grade	10B30 R SK FG RHQ 29/64 RNDCOIL
Process	DFSAR
Finish Size	0.409
Ship date	05-MAY-25

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed below and that it satisfies these requirements. The recording of false, fictitious and fraudulent statements or entries on this document may be punishable as a felony under federal statute.

NUM SPHERO=2

SPHERODIZATION (%)=86.3

CP SPHERO % LAB=0358-07

Specifications:

Manufactured per Charter Steel Quality Manual Rev Date 02/06/2025

Charter Steel certifies this product is indistinguishable from background radiation levels by having process radiation detectors in place to measure for the presence of radiation within our process & products.

Meets customer specifications with any applicable Charter Steel exceptions for the following customer documents:

Customer Document = ST 070403 rev. 14 Grade 10B30 | ST 070401 rev. 8 | SAE J403 rev. FEB2024

Revision = Dated =

Additional Comments:

Melt Source:

This MTR supersedes all previously dated MTRs for this order

Charter Steel

Saukville, WI, USA

Douglas Jones Division Mgr. of Quality Assurance

jonesdo@chartersteel.com

Trip: R6630896

Printed Date : 05/05/2025

The following statements are applicable to the material described on the front of this Test Report:

1. The laboratory that generated the analytical or test results can be identified by the following key:

Certificate Number	Lab Code	Laboratory	Address
0358-01	7388	CSSM Charter Steel Melting Division	1658 Cold Springs Road, Saukville, WI 53080
0358-02	8171	CSSR Charter Steel Rolling	1658 Cold Springs Road, Saukville, WI 53080
0358-07	8171	CSSP Charter Steel Processing Division	1658 Cold Springs Road, Saukville, WI 53080
0358-03	123633	CSFP Charter Steel Ohio Processing Division	6255 US Highway 23, Rising Sun, OH 43457
0358-04	125544	CSCM/CSCR Charter Steel Cleveland	4300 E. 49th St., Cuyahoga Heights, OH 44125-1004
*	*	---	Subcontracted test performed by laboratory not in Charter Steel System

2. When run by a Charter Steel laboratory, the following tests were performed according to the latest revisions of the specifications listed below, as noted in the Charter Steel Laboratory Quality Manual:

Test	Specifications	CSSM	CSSR	CSSP	CSFP	CSCM/CSCR
Chemistry Analysis	ASTM E415; ASTM E1019	X				X
Macroetch	ASTM E381	X				X
Hardenability (Jominy)	ASTM A255; SAE J406; JIS G0561	X				X
Grain Size	ASTM E112	X	X	X	X	X
Tensile Test	ASTM E8; ASTM A370		X	X	X	X
Rockwell Hardness	ASTM E18; ASTM A370	X	X	X	X	X
Microstructure (spheroidization)	ASTM A892		X	X	X	
Inclusion Content (Methods A, E)	ASTM E45		X			X
Decarburization	ASTM E1077		X	X	X	X

Charter Steel has been accredited to perform all of the above tests by the American Association for Laboratory Accreditation (A2LA).

These accreditations expire 01/31/27. All other test results associated with a Charter Steel laboratory that appear on the front of this report, if any, were performed according to documented procedures developed by Charter Steel and are not accredited by A2LA.

- The test results on the front of this report are the true values measured on the samples taken from the production lot. They do not apply to any other sample. Any statement of conformity is based on simple acceptance, whether the test result is within or outside the specification/acceptance limits. Measurement uncertainty is not taken into account in the statement of conformity.
- This test report cannot be reproduced or distributed except in full without the written permission of Charter Steel. The primary customer whose name and address appear on the front of this form may reproduce this test report subject to the following restrictions:
 - It may be distributed only to their customers
 - All pages must be reproduced in full
- This certification is given subject to the terms and conditions of sale provided in Charter Steel's acknowledgement (designated by our Sales Order number) to the customer's purchase order. Both order numbers appear on the front page of this Report.
- Where the customer has provided a specification, the results on the front of this test report conform to that specification unless otherwise noted on this test report.
- Certificate type 3.1 per EN 10204 rev. 2005-01
- Charter Steel melting is a continuous cast electric arc furnace
- The steel for this order meets the following compliance/ regulatory standards: DFARS, REACH, RoHS, and Conflict Minerals
- No Mercury was used in the manufacture or processing of this product.
- No weld or repair has been performed.





**CHARTER
STEEL**

A Division of
Charter Manufacturing Company, Inc.

FILE

1658 Cold Springs Road
Saukville, Wisconsin 53080
(262) 268-2400
1-800-437-8789
Fax (262) 268-2570

CHARTER STEEL TEST REPORT

Melted in USA Manufactured in USA

Brugola OEB Industriale USA
45555 Port Street
Plymouth Twp, MI-48170

Cust P.O.	4515083070
Customer Part #	9940071040
Charter Sales Order	10392533
Heat #	10912380
Ship Lot #	4884598
Grade	10B30 R SK FG RHQ 29/64 RNDCOIL
Process	DFSAR
Finish Size	0.409
Ship date	05-MAY-25

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed below and that it satisfies these requirements. The recording of false, fictitious and fraudulent statements or entries on this document may be punishable as a felony under federal statute.

Test results of Heat Lot # 10912380

Lab Code: 7388

CHEM	C	MN	P	S	SI	NI	CR	MO	CU	SN	V
%Wt	.29	.84	.007	.009	.210	.04	.08	.02	.08	.006	.003
	AL	N	B	TI	NB						
	.025	.0070	.0027	.020	.001						

JOMINY(HRC)

J1	J2	J3	J4	J5	J6	J7	J8	J9
51	50	49	46	34	24	21	20	20
J01.5MM	J03MM	J05MM	J07MM	J09MM	J11MM	J13MM	J15MM	
50	49	47	42	33	27	24	21	

JOMINY LAB=0358-01

JOMINY SAMPLE TYPE ENGLISH=R

JOMINY SAMPLE TYPE METRIC=C

E45 INCLUSION LAB=0358-02

	A	B	C	D							
Thin	1.0	1.4	.0	.6							
Heavy	.2	.2	.0	.5							
LOC1=1-6	AT=1.0	AH=.0	BT=1.5	BH=.5	CT=.0	CH=.0	DT=.5	DH=.5			
LOC2=1-11	AT=1.0	AH=.0	BT=1.0	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			
LOC3=1-12	AT=1.0	AH=.5	BT=2.0	BH=.0	CT=.0	CH=.0	DT=1.0	DH=.5			
LOC4=2-11	AT=1.0	AH=.5	BT=1.5	BH=.5	CT=.0	CH=.0	DT=.5	DH=.5			
LOC5=3-11	AT=1.0	AH=.0	BT=2.0	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			
LOC6=4-11	AT=1.0	AH=.0	BT=.5	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			

Test results of Rolling Lot # 1409828

REDUCTION RATIO=188:1

		Test results of Processing Lot #4879469, 4884598			TENSILE LAB - Metric = 0358-07 RA LAB - Metric = 0358-07
TENSILE (MPa)	# of Tests	Min Value	Max Value	Mean Value	
	1.0	586.7	586.7	586.7	
REDUCT. AREA - Metric (%)	1	62	62	62	

NUM DECARB=1

FREE FERRITE DECARB (Inch)=.000

FREE FERR & PARTIAL DECARB (Inch)=.006

Melt Source:

This MTR supersedes all previously dated MTRs for this order

Charter Steel

Saukville, WI, USA

Douglas Jones Division Mgr. of Quality Assurance

jonesdo@chartersteel.com

Trip: R6630896

Printed Date : 05/05/2025



**CHARTER
STEEL**

A Division of
Charter Manufacturing Company, Inc.

FILE

1658 Cold Springs Road
Saukville, Wisconsin 53080
(262) 268-2400
1-800-437-8789
Fax (262) 268-2570

CHARTER STEEL TEST REPORT

Melted in USA Manufactured in USA

Brugola OEB Industriale USA
45555 Port Street
Plymouth Twp, MI-48170

Cust P.O.	4515083070
Customer Part #	9940071040
Charter Sales Order	10392533
Heat #	10912380
Ship Lot #	4884598
Grade	10B30 R SK FG RHQ 29/64 RNDCOIL
Process	DFSAR
Finish Size	0.409
Ship date	05-MAY-25

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed below and that it satisfies these requirements. The recording of false, fictitious and fraudulent statements or entries on this document may be punishable as a felony under federal statute.

NUM SPHERO=1

SPHERODIZATION (%)=87.5

CP SPHERO % LAB=0358-07

Specifications:

Manufactured per Charter Steel Quality Manual Rev Date 02/06/2025

Charter Steel certifies this product is indistinguishable from background radiation levels by having process radiation detectors in place to measure for the presence of radiation within our process & products.

Meets customer specifications with any applicable Charter Steel exceptions for the following customer documents:

Customer Document = ST 070403 rev. 14 Grade 10B30 | ST 070401 rev. 8 | SAE J403 rev. FEB2024

Revision = Dated =

Additional Comments:

Melt Source:

This MTR supersedes all previously dated MTRs for this order

Charter Steel

Saukville, WI, USA

Douglas Jones Division Mgr. of Quality Assurance

jonesdo@chartersteel.com

Trip: R6630896

Printed Date : 05/05/2025

The following statements are applicable to the material described on the front of this Test Report:

1. The laboratory that generated the analytical or test results can be identified by the following key:

Certificate Number	Lab Code	Laboratory	Address
0358-01	7388	CSSM Charter Steel Melting Division	1658 Cold Springs Road, Saukville, WI 53080
0358-02	8171	CSSR Charter Steel Rolling	1658 Cold Springs Road, Saukville, WI 53080
0358-07	8171	CSSP Charter Steel Processing Division	1658 Cold Springs Road, Saukville, WI 53080
0358-03	123633	CSFP Charter Steel Ohio Processing Division	6255 US Highway 23, Rising Sun, OH 43457
0358-04	125544	CSCM/CSCR Charter Steel Cleveland	4300 E. 49th St., Cuyahoga Heights, OH 44125-1004
*	*	--	Subcontracted test performed by laboratory not in Charter Steel System

2. When run by a Charter Steel laboratory, the following tests were performed according to the latest revisions of the specifications listed below, as noted in the Charter Steel Laboratory Quality Manual:

Test	Specifications	CSSM	CSSR	CSSP	CSFP	CSCM/CSCR
Chemistry Analysis	ASTM E415; ASTM E1019	X				X
Macroetch	ASTM E381	X				X
Hardenability (Jominy)	ASTM A255; SAE J406; JIS G0561	X				X
Grain Size	ASTM E112	X	X	X	X	X
Tensile Test	ASTM E8; ASTM A370		X	X	X	X
Rockwell Hardness	ASTM E18; ASTM A370	X	X	X	X	X
Microstructure (spheroidization)	ASTM A892		X	X	X	
Inclusion Content (Methods A, E)	ASTM E45		X			X
Decarburization	ASTM E1077		X	X	X	X

Charter Steel has been accredited to perform all of the above tests by the American Association for Laboratory Accreditation (A2LA).

These accreditations expire 01/31/27. All other test results associated with a Charter Steel laboratory that appear on the front of this report, if any, were performed according to documented procedures developed by Charter Steel and are not accredited by A2LA.

3. The test results on the front of this report are the true values measured on the samples taken from the production lot. They do not apply to any other sample. Any statement of conformity is based on simple acceptance, whether the test result is within or outside the specification/acceptance limits. Measurement uncertainty is not taken into account in the statement of conformity.

4. This test report cannot be reproduced or distributed except in full without the written permission of Charter Steel. The primary customer whose name and address appear on the front of this form may reproduce this test report subject to the following restrictions:

- It may be distributed only to their customers
- All pages must be reproduced in full

5. This certification is given subject to the terms and conditions of sale provided in Charter Steel's acknowledgement (designated by our Sales Order number) to the customer's purchase order. Both order numbers appear on the front page of this Report.

6. Where the customer has provided a specification, the results on the front of this test report conform to that specification unless otherwise noted on this test report.

7. Certificate type 3.1 per EN 10204 rev. 2005-01

8. Charter Steel melting is a continuous cast electric arc furnace

9. The steel for this order meets the following compliance/ regulatory standards: DFARS, REACH, RoHS, and Conflict Minerals

10. No Mercury was used in the manufacture or processing of this product.

11. No weld or repair has been performed.



0358-01 Saukville Melting
0358-02 Saukville Rolling
0358-03 Fostoria Processing
0358-04 Cleveland Melt/Roll
0358-07 Saukville Processing



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STEEL**

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FILE

1658 Cold Springs Road
Saukville, Wisconsin 53080
(262) 268-2400
1-800-437-8789
Fax (262) 268-2570

CHARTER STEEL TEST REPORT

Melted in USA Manufactured in USA

Brugola OEB Industriale USA
45555 Port Street
Plymouth Twp, MI-48170

Cust P.O.	4515083070
Customer Part #	9940071040
Charter Sales Order	10392533
Heat #	10912380
Ship Lot #	4886492
Grade	10B30 R SK FG RHQ 29/64 RND COIL
Process	DFSAR
Finish Size	0.409
Ship date	05-MAY-25

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed below and that it satisfies these requirements. The recording of false, fictitious and fraudulent statements or entries on this document may be punishable as a felony under federal statute.

Test results of Heat Lot # 10912380

Lab Code: 7388

CHEM	C	MN	P	S	SI	NI	CR	MO	CU	SN	V
%Wt	.29	.84	.007	.009	.210	.04	.08	.02	.08	.006	.003
	AL	N	B	TI	NB						
	.025	.0070	.0027	.020	.001						

JOMINY(HRC)

J1	J2	J3	J4	J5	J6	J7	J8	J9
51	50	49	46	34	24	21	20	20
J01.5MM	J03MM	J05MM	J07MM	J09MM	J11MM	J13MM	J15MM	
50	49	47	42	33	27	24	21	

JOMINY LAB=0358-01

JOMINY SAMPLE TYPE ENGLISH=R

JOMINY SAMPLE TYPE METRIC=C

E45 INCLUSION LAB=0358-02

	A	B	C	D							
Thin	1.0	1.4	.0	.6							
Heavy	.2	.2	.0	.5							
LOC1=1-6	AT=1.0	AH=.0	BT=1.5	BH=.5	CT=.0	CH=.0	DT=.5	DH=.5			
LOC2=1-11	AT=1.0	AH=.0	BT=1.0	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			
LOC3=1-12	AT=1.0	AH=.5	BT=2.0	BH=.0	CT=.0	CH=.0	DT=1.0	DH=.5			
LOC4=2-11	AT=1.0	AH=.5	BT=1.5	BH=.5	CT=.0	CH=.0	DT=.5	DH=.5			
LOC5=3-11	AT=1.0	AH=.0	BT=2.0	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			
LOC6=4-11	AT=1.0	AH=.0	BT=.5	BH=.0	CT=.0	CH=.0	DT=.5	DH=.5			

Test results of Rolling Lot # 1409828

REDUCTION RATIO=188:1

Test results of Processing Lot #4882282, 4886492

	# of Tests	Min Value	Max Value	Mean Value	
TENSILE (MPa)	1.0	589.2	589.2	589.2	TENSILE LAB - Metric = 0358-07
REDUCT. AREA - Metric (%)	1	63	63	63	RA LAB - Metric = 0358-07

NUM DECARB=2

FREE FERRITE DECARB (Inch)=.000

FREE FERR & PARTIAL DECARB (Inch)=.004

Melt Source:

This MTR supersedes all previously dated MTRs for this order

Charter Steel

Saukville, WI, USA

Douglas Jones Division Mgr. of Quality Assurance

jonesdo@chartersteel.com

Trip: R6630896

Printed Date : 05/05/2025



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Saukville, Wisconsin 53080
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CHARTER STEEL TEST REPORT

Melted in USA Manufactured in USA

Brugola OEB Industriale USA
45555 Port Street
Plymouth Twp, MI-48170

Cust P.O.	4515083070
Customer Part #	9940071040
Charter Sales Order	10392533
Heat #	10912380
Ship Lot #	4886492
Grade	10B30 R SK FG RHQ 29/64 RNDCOIL
Process	DFSAR
Finish Size	0.409
Ship date	05-MAY-25

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed below and that it satisfies these requirements. The recording of false, fictitious and fraudulent statements or entries on this document may be punishable as a felony under federal statute.

NUM SPHERO=2

SPHERODIZATION (%)=86.3

CP SPHERO % LAB=0358-07

Specifications:

Manufactured per Charter Steel Quality Manual Rev Date 02/06/2025

Charter Steel certifies this product is indistinguishable from background radiation levels by having process radiation detectors in place to measure for the presence of radiation within our process & products.

Meets customer specifications with any applicable Charter Steel exceptions for the following customer documents:

Customer Document = ST 070403 rev. 14 Grade 10B30 | ST 070401 rev. 8 | SAE J403 rev. FEB2024

Revision = Dated =

Additional Comments:

Melt Source:

Charter Steel

Saukville, WI, USA

This MTR supersedes all previously dated MTRs for this order

Douglas Jones Division Mgr. of Quality Assurance

jonesdo@chartersteel.com

Trip: R6630896

Printed Date : 05/05/2025

The following statements are applicable to the material described on the front of this Test Report:

1. The laboratory that generated the analytical or test results can be identified by the following key:

Certificate Number	Lab Code	Laboratory	Address
0358-01	7388	CSSM Charter Steel Melting Division	1658 Cold Springs Road, Saukville, WI 53080
0358-02	8171	CSSR Charter Steel Rolling	1658 Cold Springs Road, Saukville, WI 53080
0358-07	8171	CSSP Charter Steel Processing Division	1658 Cold Springs Road, Saukville, WI 53080
0358-03	123633	CSFP Charter Steel Ohio Processing Division	6255 US Highway 23, Rising Sun, OH 43457
0358-04	125544	CSCM/CSCR Charter Steel Cleveland	4300 E. 49th St., Cuyahoga Heights, OH 44125-1004
*	*	--	Subcontracted test performed by laboratory not in Charter Steel System

2. When run by a Charter Steel laboratory, the following tests were performed according to the latest revisions of the specifications listed below, as noted in the Charter Steel Laboratory Quality Manual:

Test	Specifications	CSSM	CSSR	CSSP	CSFP	CSCM/CSCR
Chemistry Analysis	ASTM E415; ASTM E1019	X				X
Macroetch	ASTM E381	X				X
Hardenability (Jominy)	ASTM A255; SAE J406; JIS G0561	X				X
Grain Size	ASTM E112	X	X	X	X	X
Tensile Test	ASTM E8; ASTM A370		X	X	X	X
Rockwell Hardness	ASTM E18; ASTM A370	X	X	X	X	X
Microstructure (spheroidization)	ASTM A892		X	X	X	
Inclusion Content (Methods A, E)	ASTM E45		X			X
Decarburization	ASTM E1077		X	X	X	X

Charter Steel has been accredited to perform all of the above tests by the American Association for Laboratory Accreditation (A2LA).

These accreditations expire 01/31/27. All other test results associated with a Charter Steel laboratory that appear on the front of this report, if any, were performed according to documented procedures developed by Charter Steel and are not accredited by A2LA.

- The test results on the front of this report are the true values measured on the samples taken from the production lot. They do not apply to any other sample. Any statement of conformity is based on simple acceptance, whether the test result is within or outside the specification/acceptance limits. Measurement uncertainty is not taken into account in the statement of conformity.
- This test report cannot be reproduced or distributed except in full without the written permission of Charter Steel. The primary customer whose name and address appear on the front of this form may reproduce this test report subject to the following restrictions:
 - It may be distributed only to their customers
 - All pages must be reproduced in full
- This certification is given subject to the terms and conditions of sale provided in Charter Steel's acknowledgement (designated by our Sales Order number) to the customer's purchase order. Both order numbers appear on the front page of this Report.
- Where the customer has provided a specification, the results on the front of this test report conform to that specification unless otherwise noted on this test report.
- Certificate type 3.1 per EN 10204 rev. 2005-01
- Charter Steel melting is a continuous cast electric arc furnace
- The steel for this order meets the following compliance/ regulatory standards: DFARS, REACH, RoHS, and Conflict Minerals
- No Mercury was used in the manufacture or processing of this product.
- No weld or repair has been performed.

